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Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 16: Organisational Structures

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

The lecture addressed:

- The different concepts of organisational metrications including: Key performance indicators (KPIs), PIs and Metrics were discussed.
- Some Facts about KPIs, Anatomy of a KPI, Attributes and creation of a good KPI, KPIs often used in Project Management and KPIs identification in the semester project were all discussed.

Learning Outcomes for the Current Lecture

At the end of this lecture you should be able to:

- **Define, discuss, and analyse** an organisation's structure and organisational work flow drivers
- **Discuss, analyse and evaluate** types, attributes, Pros and Cons. of organisational structures, covering:
 - Traditional (Classical) organisation structure
 - Product organisation structure
 - Matrix organisation structure
- **Apply** this concept to the semester project (Not part of the report)

Presentation Outline

- Definition of an Organisational Structure
- Organisational Work Flow Drivers
- Types and Attributes of Organisation Structures
 - Traditional (Classical) Organisation Structure
 - Product Organisation Structure
 - Matrix Organisation Structure

Definition of an Organisational Structure

- An organisation structure is an integrated operational hierarchy within an enterprise. It identifies each job, its function/s and where it reports to within the structure.
- This structure is developed to establish how an organisation operates and assists in obtaining its goals to allow for future growth.
- The structure is often illustrated using an organisational chart.

Definition of an Organisation Structure

Organisation structure determines how the roles or responsibilities and power are assigned, controlled, and coordinated, and how information flows across the different levels of management.

It depicts the typical hierarchical arrangement of lines of authorities, communications, rights and duties in a formal set up.

Key Attributes in the Definition of an Organisation Structure:

- **Authority** is the power granted to individuals (possibly by their position) so that they can make final decisions.
- **Responsibility** is the obligation incurred by individuals in their roles in the formal organisation to effectively perform assignments.
- **Accountability** is being answerable for the satisfactory completion of a specific assignment. (Accountability = authority + responsibility)

Need for Organisation Structure/Work Flow Drivers

Organisations are continually being restructured to meet the demands imposed by the environment. Restructuring can change the role of individuals in the formal and the informal organisation. Whichever organisation structure that is finally selected, individual team members must have a clear description of the authority, responsibility, and accountability accrued to them for the work to proceed.

Need for Organisation Structure/Work Flow Drivers

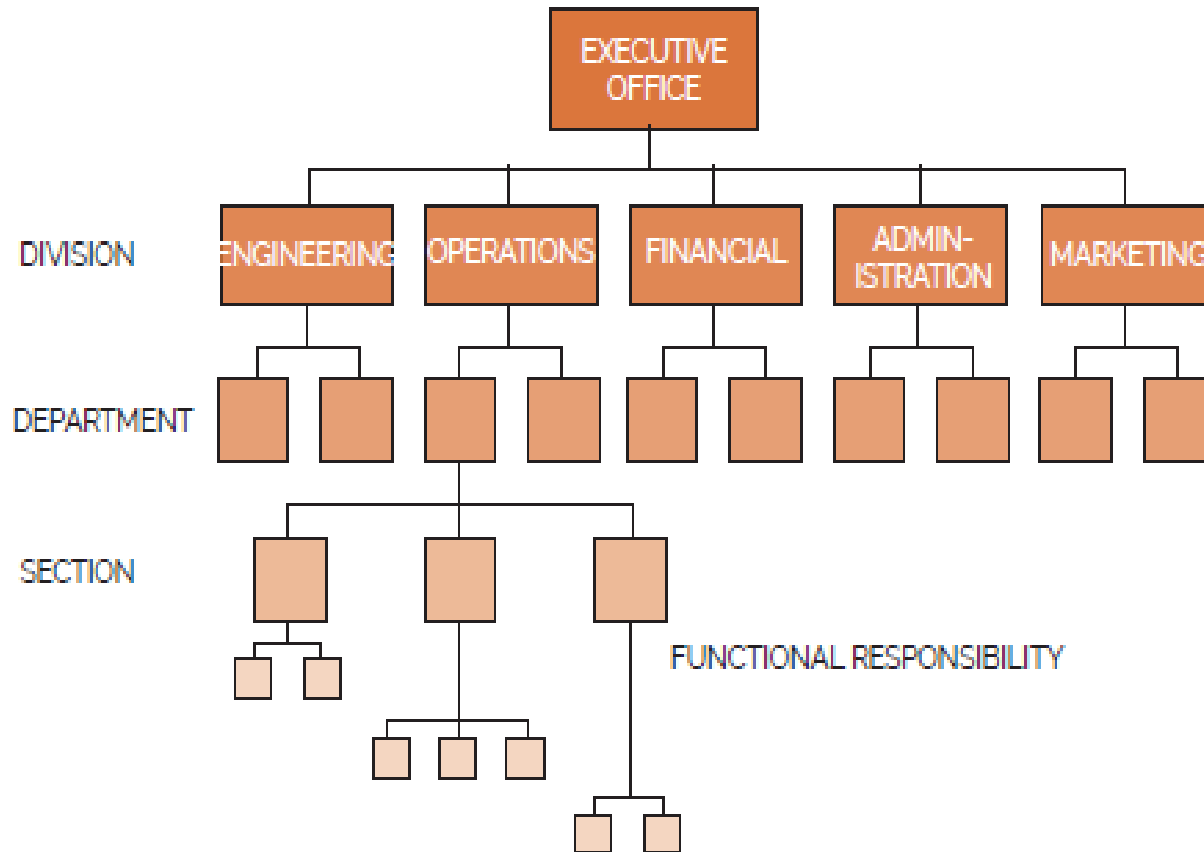
Organisational work flow are often structured/designed based on one or more of the following drivers: **Work Divisions, Task Functions, Product, Geographical Location, Complexity, Need for Efficiency, Collaboration** amongst others.

A divisional structure is suitable for organisations with distinct business units, while a geographical structure provides a hierarchy for organisations that operate at several locations nationally or internationally. A functional organisational structure is based on each job's duties while collaborative or matrix flow often have two or several supervisors for each job to report to.

Types, Attributes, Pros & Cons of Organisational Structures

Traditional (Classical) organisation

The traditional management structure has survived for more than two centuries. However, recent business developments, such as the rapid rate of change in technology and increased demands of stakeholders, have created strains on existing organisational forms. Several years ago, companies could survive with only one or two product lines however, that may not be too successful in these days due to the extreme customised and dynamic environment and population explosion.



Traditional Management Structure

Advantages of The Traditional (Classical) organisation

- Easier budgeting and cost control are possible.
- Better technical control is possible.
 - Specialists can be grouped to share knowledge and responsibility.
 - Personnel can be used on many different projects.
 - All projects will benefit from the most advanced technology (better utilization of scarce personnel).
- Flexibility in the use of manpower.
- A broad manpower base to work with.
- Continuity in the functional disciplines; policies, procedures, and lines of responsibility are easily defined and understandable

Advantages of The Traditional (Classical) organisation

- Admits mass production activities within established specifications.
- Good control over personnel, since each employee has one and only one person to report to.
- Communication channels are vertical and well established.
- Quick reaction capability exists, but may be dependent upon the priorities of the functional managers.

Disadvantages of The Traditional (Classical) organisation

- No one individual is directly responsible for the total project (i.e., no formal authority; committee solutions).
- Does not provide the project-oriented emphasis necessary to accomplish the project tasks.
- Coordination becomes complex, and additional lead time is required for approval of decisions.
- Decisions normally favour the strongest functional groups.
- No customer focal point.
- Response to customer needs is slow.

Disadvantages of The Traditional (Classical) organisation

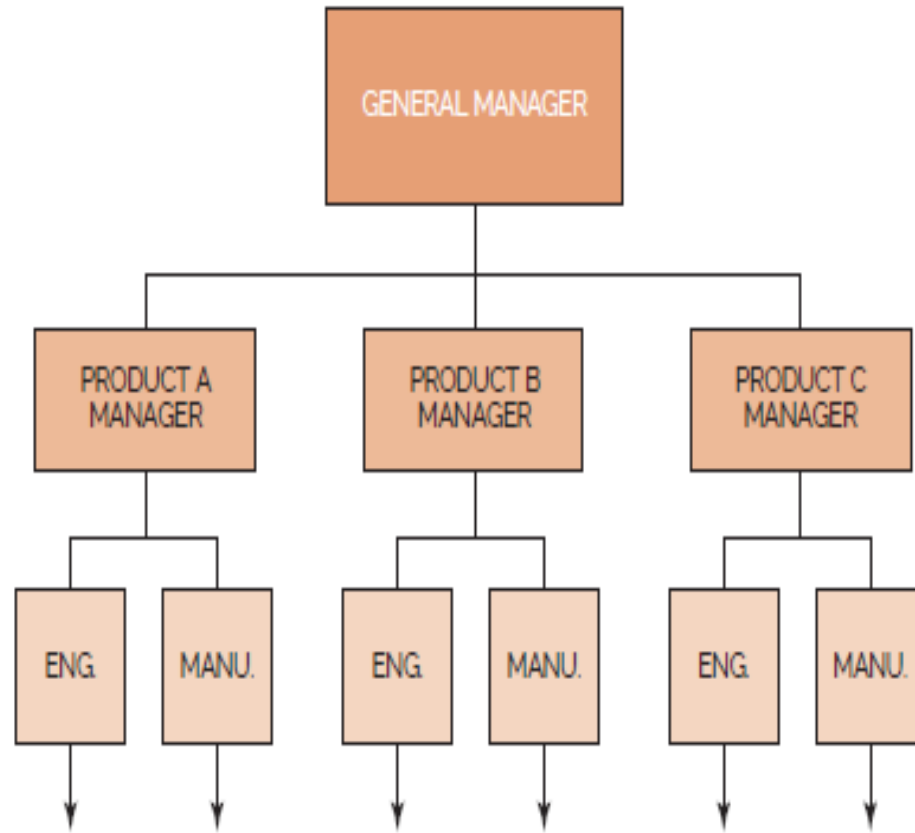
- Difficulty in pinpointing responsibility; this is the result of little or no direct project reporting, very little project-oriented planning, and no project authority.
- Motivation and innovation are decreased.
- Ideas tend to be functionally oriented with little regard for ongoing projects.

Product Organisation Structure

A product based organisation is one that is structured around a specific product. It's often referred to as a division within a division. It could be likened to a situation of having different team members working on the same type of products for different clients or different types of products for different clients.

Product Organisation Structure

With the passing of time, companies found that survival depended on multiple product lines (i.e., diversification) and vigorous integration of technology into the existing organisation. As organisations grew and matured, managers found that company activities were not being integrated effectively, and that new conflicts were arising in the well-established formal and informal channels. Managers began searching for more innovative organisational forms that would alleviate these problems.



Product Organisational Structure

Advantages of a Product Organisational Structure

- Provides complete line authority over the project (i.e., strong control through a single project authority).
- Participants work directly for the project manager. Unprofitable product lines are easily identified and can be eliminated.
- Strong communications channels.
 - Staffers can maintain expertise on a given project without sharing key personnel.
 - Very rapid reaction time is provided.
 - Personnel demonstrate loyalty to the project; better morale with product identification.

Advantages of a Product Organisation Structure

- A focal point develops for out-of-company customer relations.
- Flexibility in determining time (schedule), cost, and performance trade-offs.
- Interface management becomes easier as unit size is decreased.
- Upper-level management maintains more free time for executive decision making.

Disadvantages of a Product Organisation Structure

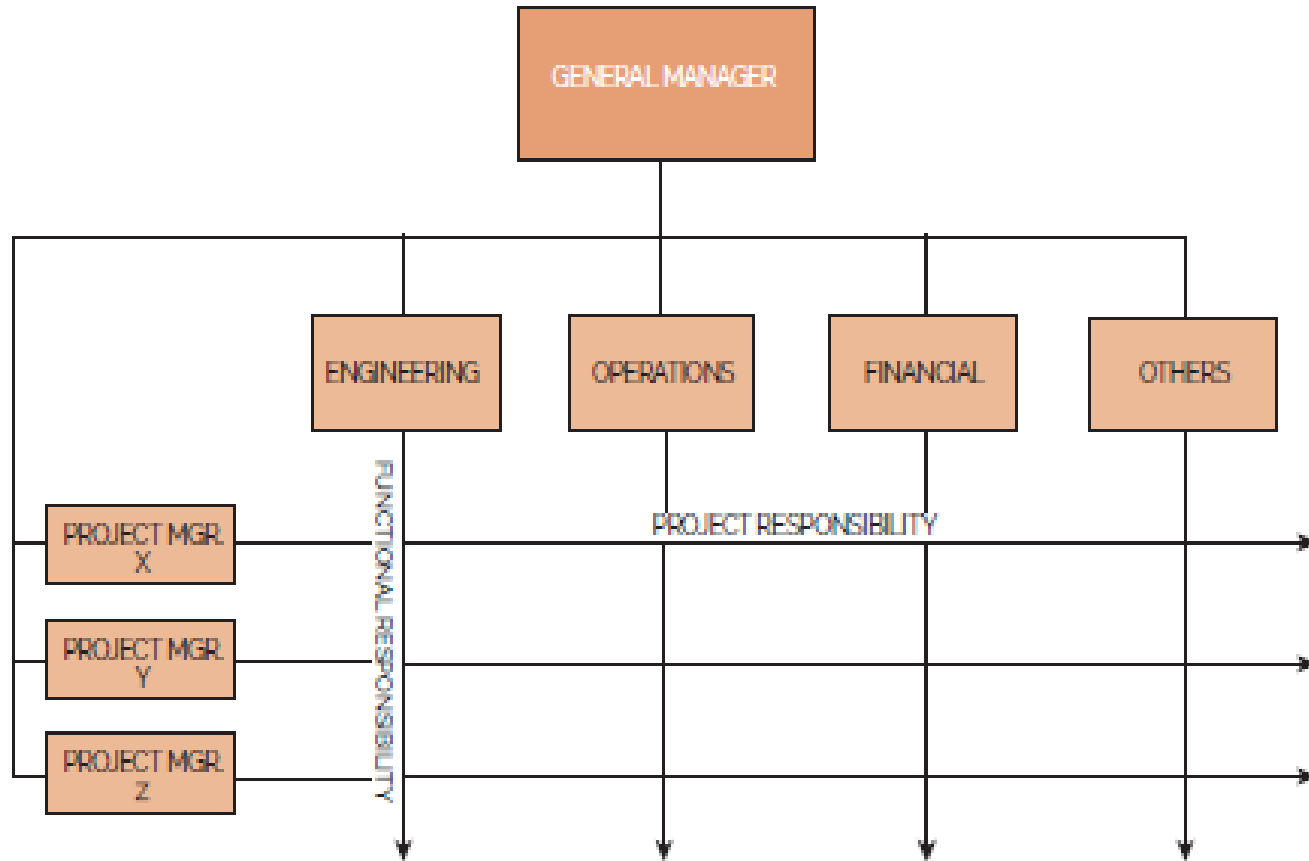
- Cost of maintaining this form in a multiproduct company would be high due to duplication of effort, facilities, and personnel.
- A tendency to retain personnel on a project long after they are needed may decline.
- Technology suffers because, without strong functional groups, outlook of the future to improve company's capabilities for new programs would be hampered (i.e., no perpetuation of technology).

Disadvantages of a Product Organisation Structure

- Control of functional (i.e., organisational) specialists requires top-level coordination.
- Lack of opportunities for technical interchange between projects.
- Lack of career continuity and opportunities for project personnel.

Matrix Organisation Structure

The matrix organisation form is an attempt to combine the advantages of the pure functional structure and the product organisational structure. This form is ideally suited for “project-driven” companies. The advantages of the matrix structure eliminates almost all of the disadvantages of the traditional structure.



Typical matrix structure

Matrix Organisational Structure

The word “matrix” often brings fear to the hearts of executives because it implies radical change, or at least they think that it does. The matrix is simply horizontal lines superimposed over the traditional structure. The horizontal lines will come and go as projects start up and terminate, but the traditional structure will remain.

Matrix Organisation Structure

- How can a functional unit (operating in a vertical structure) answer questions and achieve project goals and objectives that are compatible with other projects?
- The answers to these questions depend on mutual understanding between the project and functional managers. Since both individuals maintain some degree of authority, responsibility, and accountability on each project, they must continuously negotiate.

Matrix Organisational Structure

The project environment and functional environment cannot be separated but must interact. The location of the project and functional unit interface is the focal point for all activities. The functional manager controls departmental resources (i.e., people). This poses a problem because, although the project manager maintains the maximum control (through the line managers) over all resources including cost and personnel, the functional manager must provide staff for the project's requirements.

Matrix Organisation Structure

Rules for matrix development:

- Participants are full time on the project; this ensures a degree of loyalty.
- Horizontal as well as vertical channels must exist for making commitments.
- There must be quick and effective methods for conflict resolution.
- There must be good communication channels and free access between managers.
- All managers must have input into the planning process.
- Both horizontally and vertically oriented managers must be willing to negotiate for resources.
- The horizontal line must be permitted to operate as a separate entity except for administrative purposes.

Advantages of the Matrix Organisation Structure

- The project manager maintains maximum project control (through the line managers) over all resources, including cost and personnel.
- Policies and procedures can be set up independently for each project, provided that they do not contradict company policies and procedures.
- The project manager has the authority to commit company resources, provided that scheduling does not cause conflicts with other projects.

Advantages of the Matrix Organisation Structure

- Rapid responses are possible to changes, conflict resolution, and project needs (as technology or schedule).
- The functional organisations exist primarily as support for the project.
- Each person has a “home” after project completion. Each person can be shown a career path.

Advantages of the Matrix Organisation Structure

- Because key people can be shared, the program cost is minimized. People can work on a variety of problems; that is, better people control is possible.
- A strong technical base can be developed, and much more time can be devoted to complex problem solving. Knowledge is available for all projects on an equal basis.
- Conflicts are minimal, and those requiring hierarchical referrals are more easily resolved.

Advantages of the Matrix Organisation Structure

- There is a better balance among time, cost, and performance.
- Rapid development of specialists and generalists also exists.
- Authority and responsibility are shared.
- Stress is distributed among the team (and the functional managers).
- Multidimensional information flow.
- Multidimensional workflow.
- Dual reporting.
- Continuously changing priorities.

Disadvantages of the Matrix Organisation Structure

- Management goals different from project goals.
- Potential for continuous conflict and conflict resolution.
- Difficulty in monitoring and control.
- Company-wide, the organisational structure is not cost-effective because more people than necessary are required, primarily administrative.
- Each project organisation operates independently. Care must be taken that duplication of efforts does not occur.

Disadvantages of the Matrix Organisation Structure

- More effort and time are needed initially to define policies and procedures, compared to the traditional form.
- Functional managers may be biased according to their own set of priorities.
- Balance of power between functional and project organisations must be watched.
- Balance of time, cost, and performance must be monitored.
- Although rapid response time is possible for individual problem resolution, the reaction time can become quite slow.
- Employees and managers are more susceptible to role ambiguity than in a traditional form.

Disadvantages of the Matrix Organisation Structure

- Conflicts and their resolution may be a continuous process (possibly requiring support of an organisational development specialist).
- People do not feel that they have any control over their own destiny when continuously reporting to multiple managers.

Lecture Resources

Recommended readings:

- Project Management - A Multi-disciplinary Approach (4th edition)
Chapter 3: Organisational Structures

<https://smallbusiness.chron.com/meaning-organizational-structure-3803.html>

<http://www.businessdictionary.com/definition/organizational-structure.html>

Learning Outcomes Evaluated

- Are you able to define, discuss, and analyse an organisation's structure and organisational workflow drivers
- Can you analyse and evaluate types, attributes, Pros and Cons. of organisational structures, with emphasis on the:
 - Traditional (Classical) organisation structure
 - Product organisation structure
 - Matrix organisation structure?
- Assuming you are meant to apply this concept to the semester project, which structure will you approve for your business?

The End



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